

Historically, Bow shock and IP shock often analysed by different sub fields

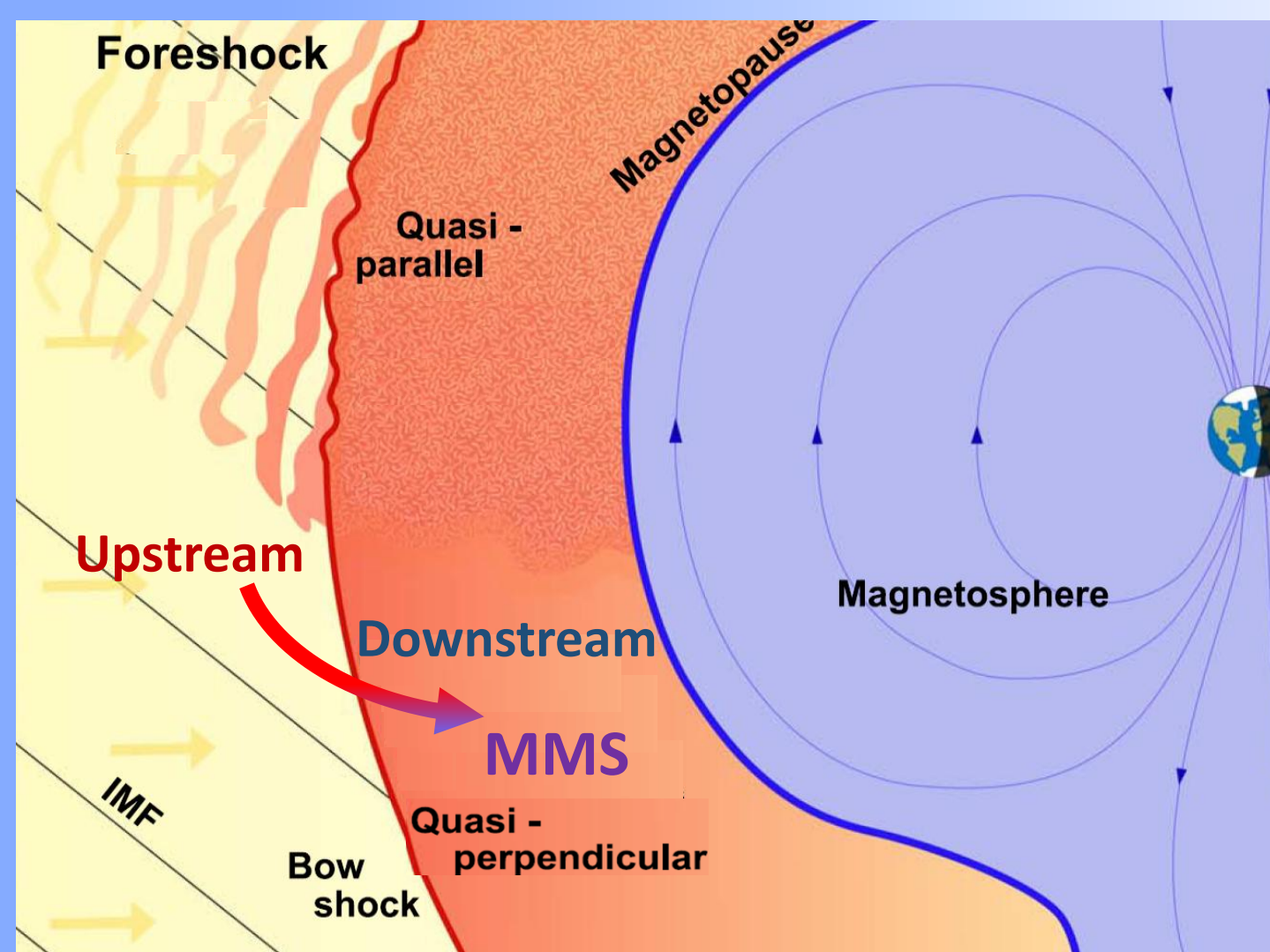
Collisionless Shocks are ubiquitous and universal.
A systematic comparative picture across Mach number regimes is still lacking.
Can we use Measurements at Earth's bow shock to cross-compare with IP shocks?

Required:
Get shocks with similar strength (Mach number) for cross-comparisons

Collisionless Shocks

Planetary Bow Shocks: Strong Shocks

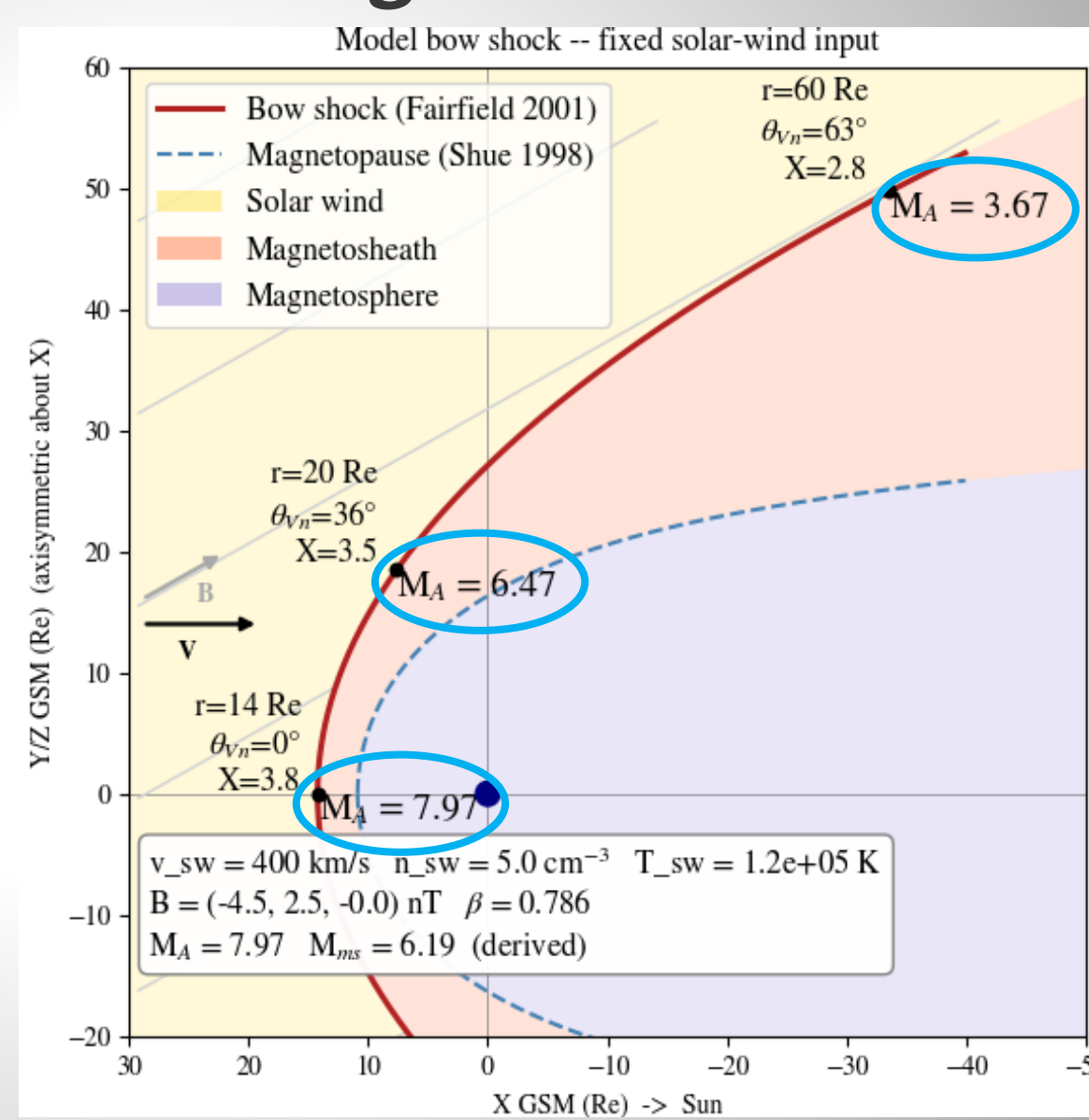
Planetary magnetic fields are obstacles for the solar wind: the solar wind plasma gets suddenly decelerated, compressed, and heated.



Flank Bow Shock Crossings: Weak Shocks

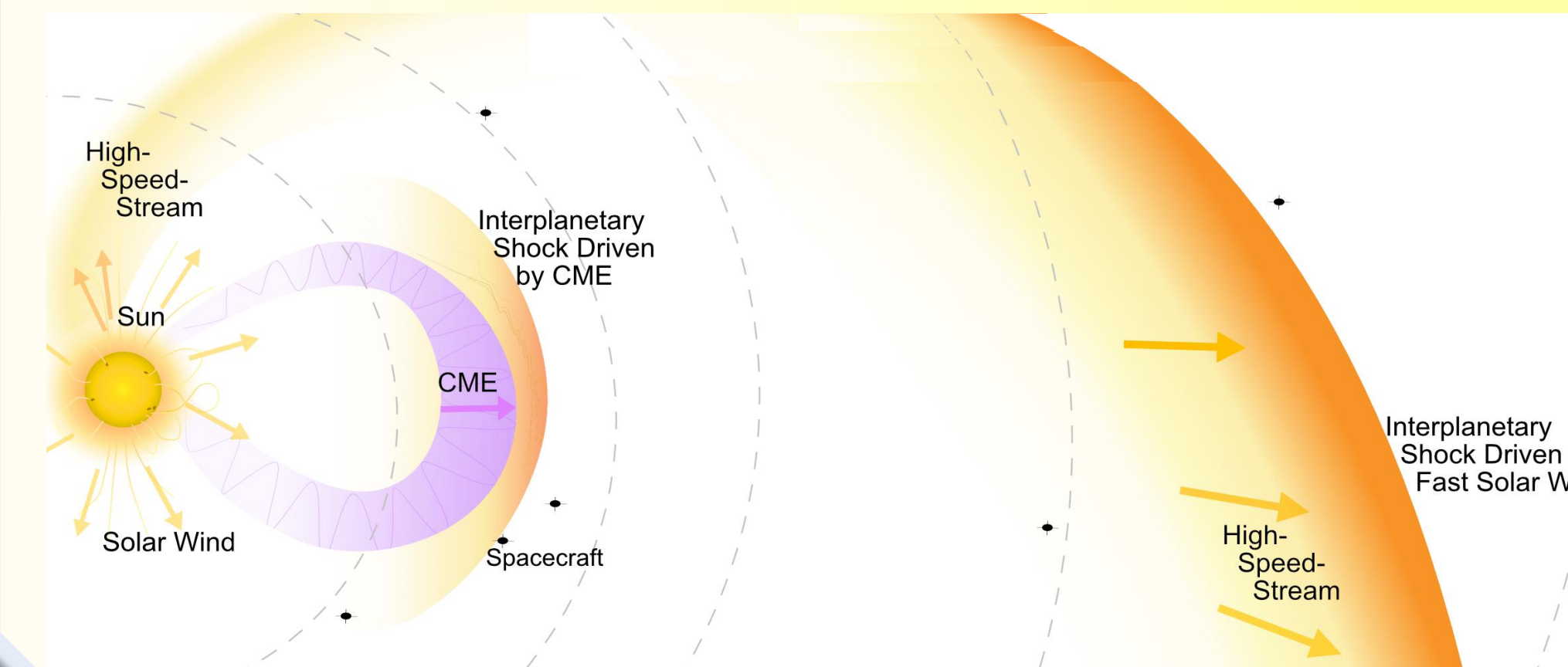
The velocity **perpendicular to the shock surface** gets decelerated and determines shock strength.
At the flank of a planetary bow shock, the incident angle is low.

Therefore, the flank shock is weaker.



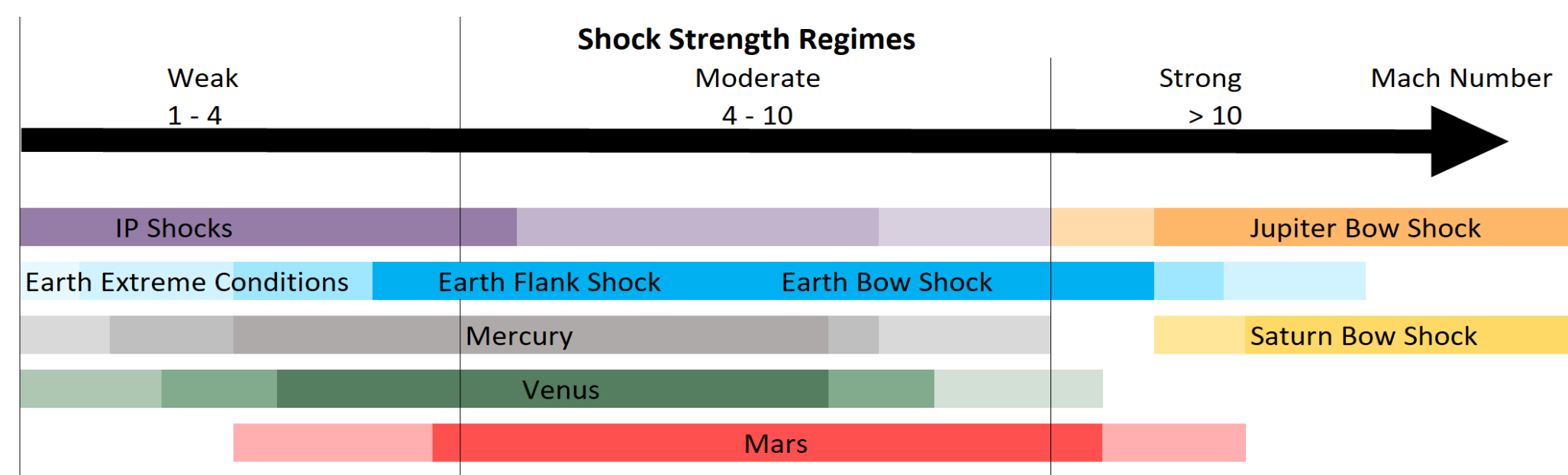
Interplanetary Shocks: Weak Shocks

Faster solar wind (caused by CMEs or coronal holes) catches up with slow wind. Piled up plasma causes interplanetary shock, moving with the solar wind through the solar system.



Overview of Shock Lists

The IP Shocks Database (Kilpua+2026): <https://ipshocks.helsinki.fi/>
SHARP Database (Ganushkina+2024): <https://sharp.fmi.fi/shock-database/>
CFA IP Shock Database: <https://web.cfa.harvard.edu/shocks/>
IP Shock List (Oliveira 2023): <https://zenodo.org/records/7991430>
MMS Burst Mode Shocks (Toy-Edens+2025): <https://zenodo.org/records/15318626>
Bow Shock Crossing Database (covering 1973-2002): <https://omniweb.gsfc.nasa.gov/ftpbrowser/bowshock.html>
SERPENTINE Shock Catalogues: <https://data.serpentine-h2020.eu/>
Wind Bow Shock List (1994-2003): https://wind.nasa.gov/mfi/bow_shock.html



Relevant Parameters for Mach Numbers:

- Local plasma information speeds
- Shock Angle Theta Bn (orientation of B and shock surface)
- (normal to shock) Velocity (upstream velocity and shock orientation)
- Upstream Plasma Beta <highly correlated; relevant for shock criticality> (upstream B and thermal pressure)

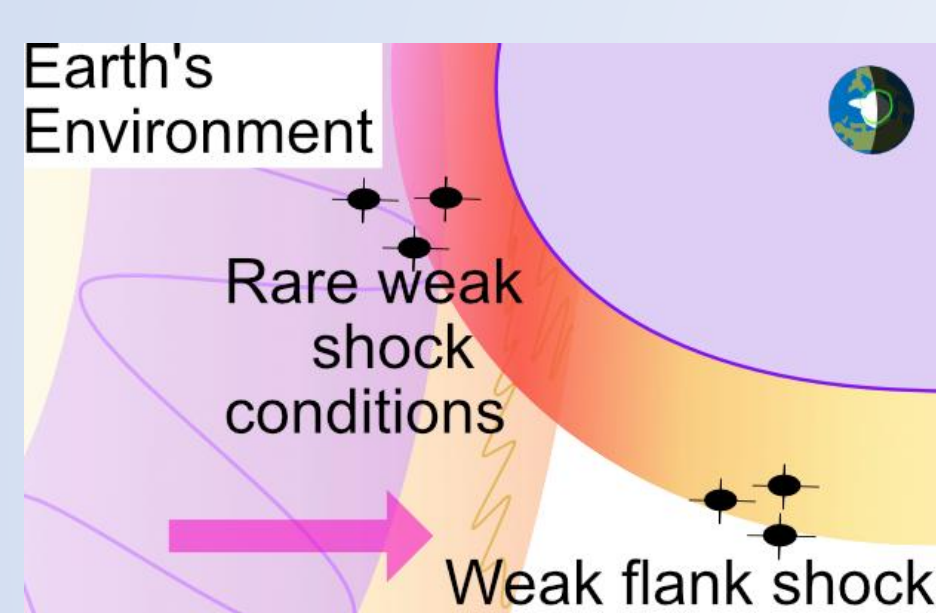
$$M_A = \frac{V_u}{v_A} = \frac{V_u}{B_u / (\mu_0 n_u m_i)^{1/2}}$$

Alfvén Mach number M_A often preferred over fast magnetosonic Mach number (simpler, minimal measurement constraints)

Weak terrestrial bow shock crossings are within CMEs.
Earth's bow shock is weak during low solar wind density and very high magnetic field strength: conditions found in CMEs!

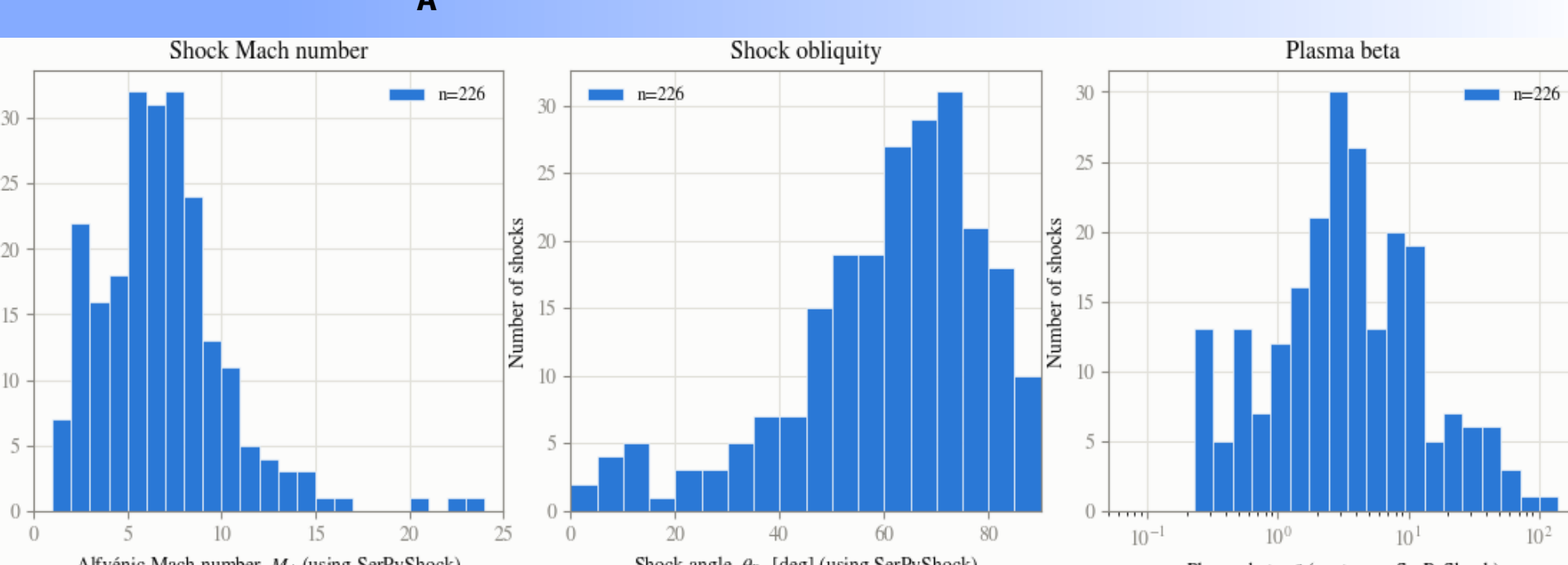
Caveat: Low β and high alpha particle abundance affects plasma behaviour.

Can we still compare conditions with IP shocks?



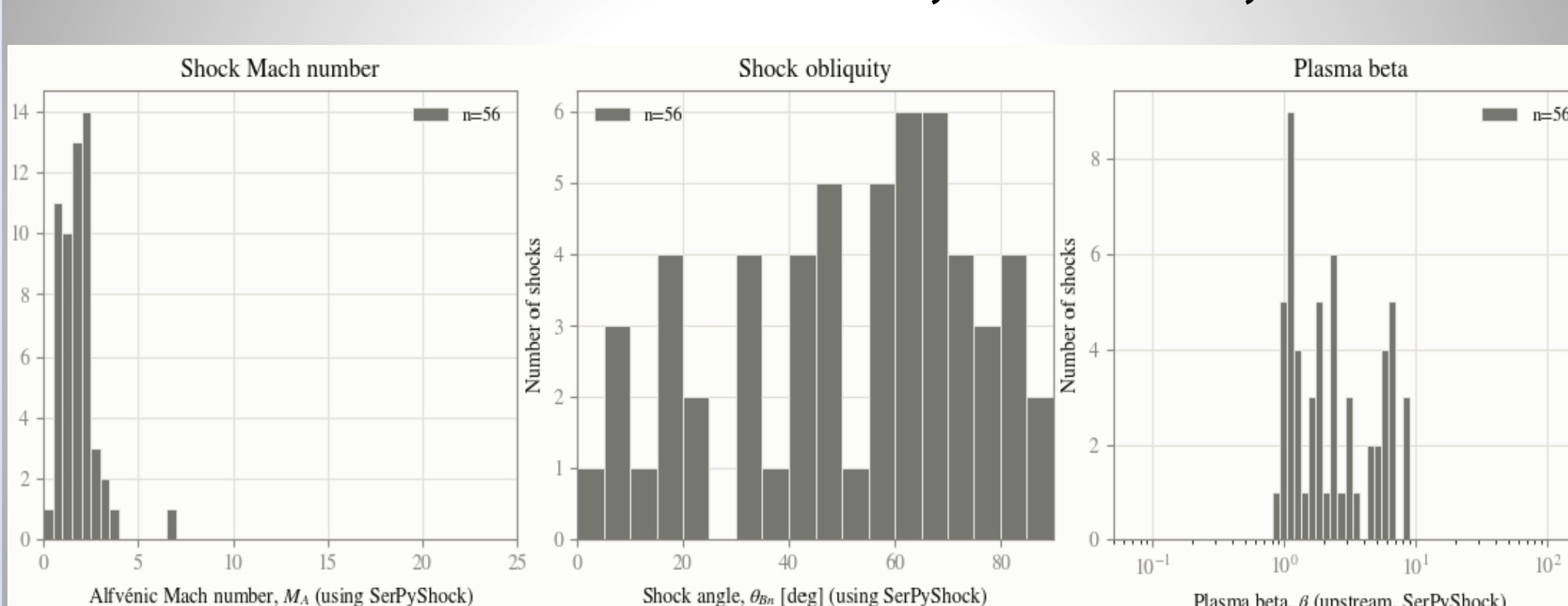
CMEs create lowest shock Mach numbers and highest pressure anisotropies!

Burst Data Shock Statistics with MMS 2015 – 2023 (and expanding) With added Low M_A events. n=226



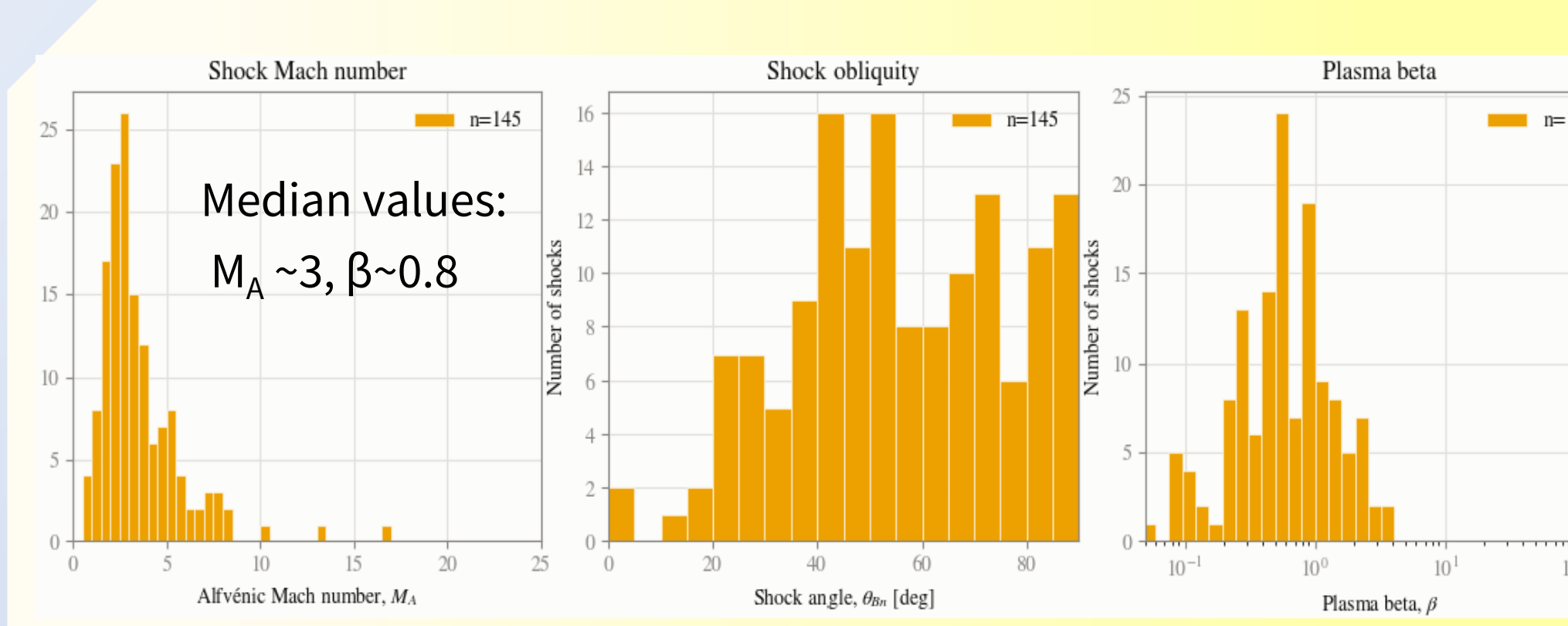
Median values: $M_A \sim 7$, $\beta \sim 3$
All events with $M_A < 3$ are due to strong CME events

Flank Shock Statistics with ARTEMIS (THEMIS B & C) in orbit around moon at ~ 60 Re distance, self made list, n=56



Median values: $M_A \sim 2$, $\beta \sim 2$

Selected IP Shock Statistics (Solar Orbiter, Serpentine), n=137



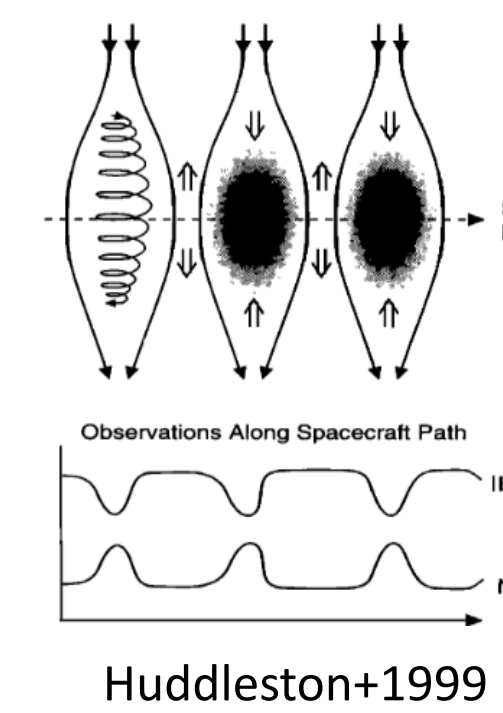
Due to high velocity relative to a spacecraft (400km/s +) the available amount of data during the shock crossing is very limited. High resolution solar wind sampling is needed!

Statistical Example Analysis: Ion Plasma Instability Downstream of Quasi-Perpendicular Shocks

Quasi-perpendicular shocks cause **temperature anisotropy increase!**
Ions perpendicular to magnetic field heat up more.

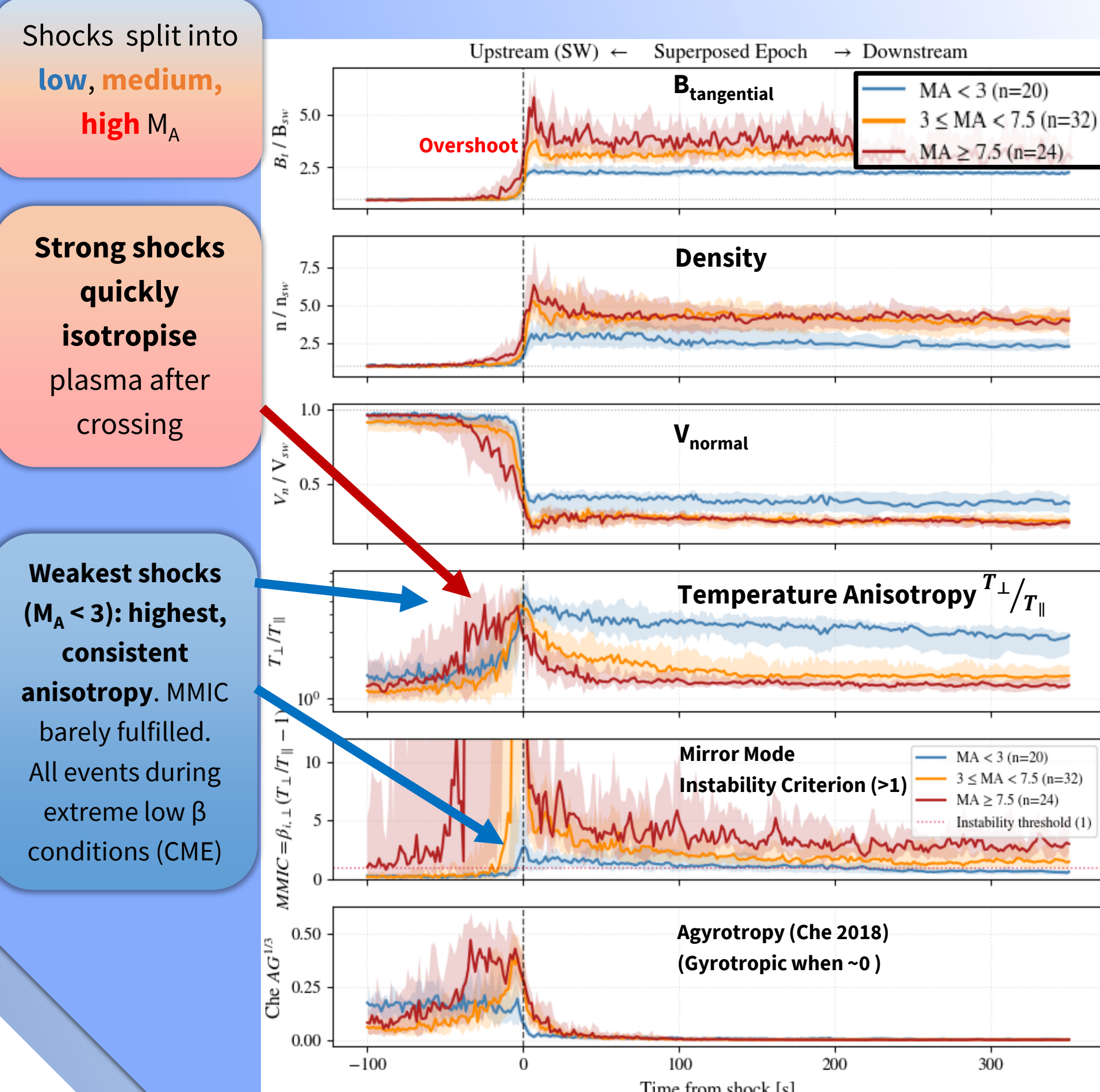
Temperature anisotropy ($T_{\perp}/T_{\parallel}$) – Ratio of pressure along and perpendicular to magnetic field line.

-> **Instabilities (like mirror modes) are plasma's way to isotropise again**



Earth Bow Shock: Superposed Epoch Analysis

Events are enriched with upstream data (OMNI, Wind), upstream fluctuations, alpha/proton abundances, and all shock parameters were estimated using Serpyshock (Trotta et al. 2022). Only q-perp

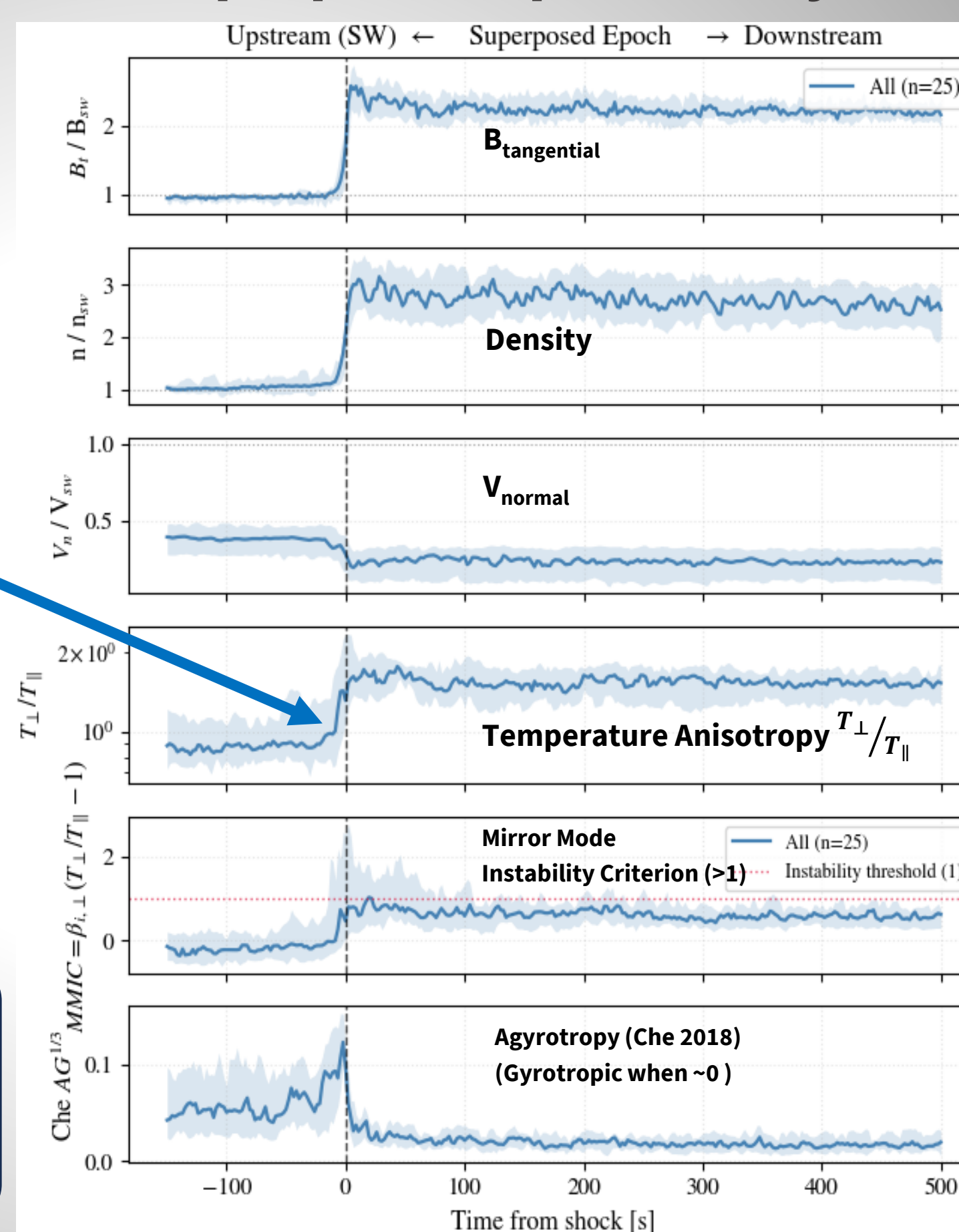


Earth's Flank Shocks: Superposed Epoch Analysis

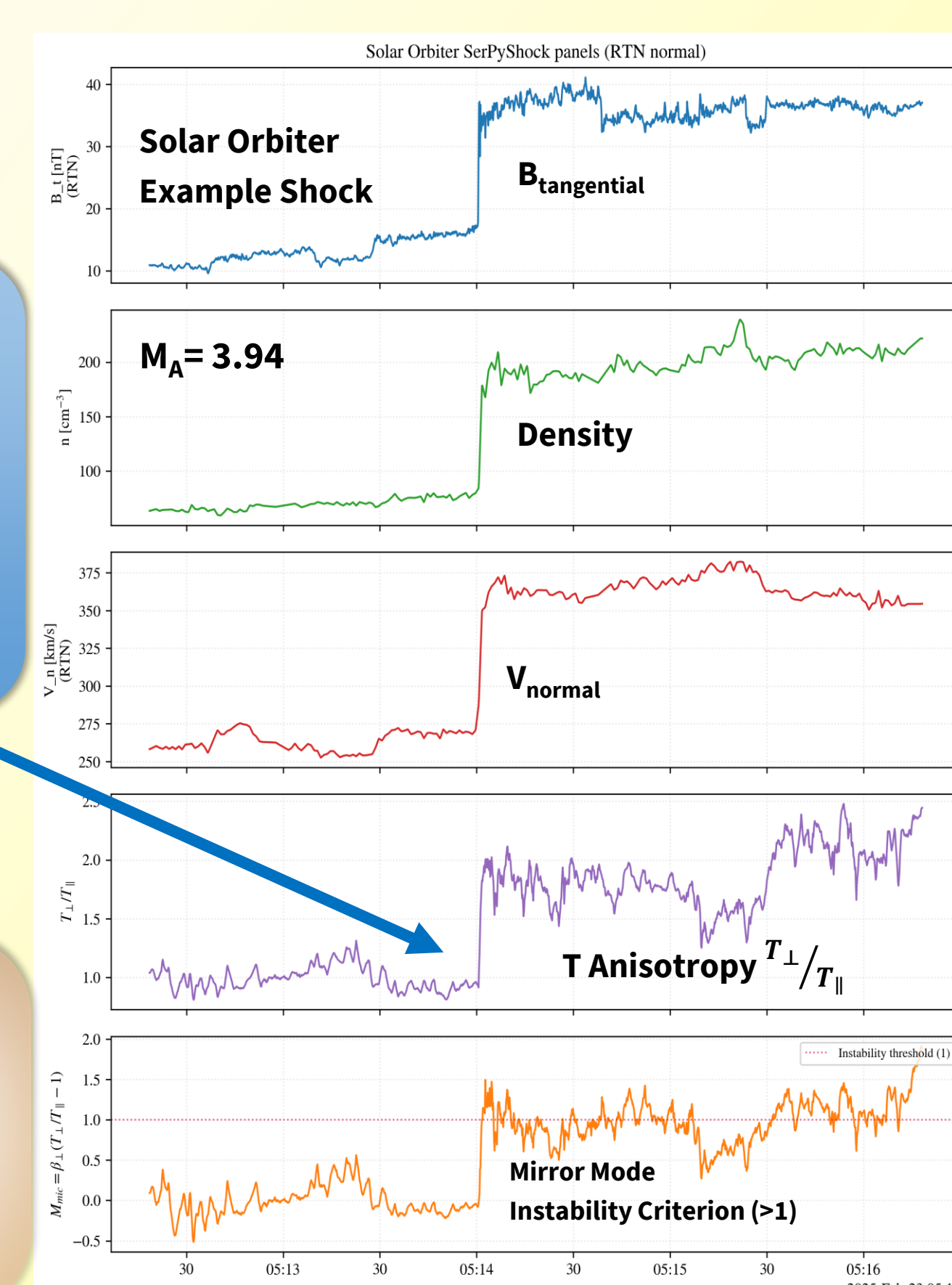
Events enriched with upstream (OMNI, Wind) information, all shock parameters estimated using Serpyshock (Trotta et al. 2022). Only q-perp shocks.

Weak shock at flanks show **modest $T_{\perp}/T_{\parallel}$ increase** and barely isotropisation. (plasma is still in stable conditions, ask me for the Brazil plots)

Flank Shocks provide very weak shock conditions with no plasma beta conditioning - comparable to IP shocks?



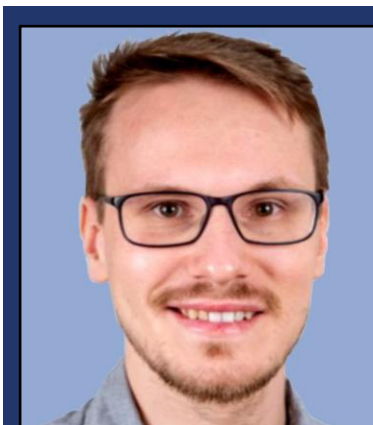
Interplanetary Shocks: WORK IN PROGRESS



Similar $T_{\perp}/T_{\parallel}$ increase, but little time close to shock (shock speed ~ 400 km/s)

Will statistics hold the same relation as the bow shock results?

IP shocks: Difference in size and geometry (neglectable curvature), and very limited observation duration close to the actual shock



Corresponding author:
f.koller@qmul.ac.uk

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